

Berker Demirel

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EDUCATION

- **Institute of Science and Technology Austria** Vienna, Austria
PhD Candidate in Computer Science Sep 2023 - Current
- **Sabanci University** Istanbul, Turkey
Masters in Computer Science; GPA: 3.91 Sep 2021 - June 2023
- **Sabanci University** Istanbul, Turkey
BS Computer Science and Engineering, BA Economics (DM) and Mathematics (Minor); GPA: 3.84 Sep 2015 - Jun 2021

SKILLS SUMMARY

- **Languages:** Python, C++, C, Unix scripting
- **Tools:** PyTorch, git, OpenMP, CUDA, CPLEX

SELECTED PUBLICATIONS

- Berker Demirel, Valentino Maiorca, Marco Fumero, Theofanis Karaletsos and Francesco Locatello, **Assessing Sample Quality in Conditional Generation under Compositional Shift**, arXiv preprint arXiv:2606.09601, 2026.
- Berker Demirel, Marco Fumero, Theofanis Karaletsos and Francesco Locatello, **MorphGen: Controllable and Morphologically Plausible Generative Cell-Imaging**, arXiv preprint arXiv:2510.01298, 2025.
- Berker Demirel, Marco Fumero and Francesco Locatello, **Out-of-Distribution Detection with Relative Angles**, *Advances in Neural Information Processing Systems* 38 (2025).
- Berker Demirel, Lingjing Kong, Kun Zhang, Theofanis Karaletsos, Celestine Mendler-Dünnier and Francesco Locatello, **Adjusting Pretrained Backbones for Performativity**, arXiv preprint arXiv:2410.04499, 2024.
- Berker Demirel and Huseyin Ozkan, **Decompl: Decompositional learning with attention pooling for group activity recognition from a single volleyball image**, in 2024 IEEE International Conference on Image Processing (ICIP). IEEE.
- Berker Demirel, Erchan Aptoula and Huseyin Ozkan, **ADRMX: Additive Disentanglement of Domain Features with Remix Loss**, arXiv preprint arXiv:2308.06624, 2023.

RESEARCH EXPERIENCE

- **ICHEC (Irish Centre for High End Computing)** Dublin, Ireland
Research Intern - Prof. Buket Benek Gursoy Jun 2020 - Aug 2020
 - Participated in the project “GPU Acceleration of Breadth-First Search algorithm in applications of Social Networks” for two months.
 - Proposed 7 different Direction Optimized BFS algorithms for CUDA and created a benchmark.
- **Erasmus University Rotterdam** Rotterdam, Netherlands
Research Intern - Prof. Ilker Birbil Jun 2019 - Aug 2019
 - Implemented Policy-Gradient based and Value based Reinforcement Learning Algorithms to solve Graph Coloring Problem
 - Prepared a technical report about proposed algorithms and presented the study to the supervisor

TEACHING EXPERIENCE

- **Institute of Science and Technology Austria** Klosterneuburg, Austria
Teaching Assistant Sep 2024 - Current
 - **Course:** Modern Machine Learning
 - Holding weekly recitation hours
 - Preparing and grading homeworks.
- **Sabanci University** Istanbul, Turkey
Teaching Assistant Sep 2021 - 2023
 - **Courses:** Machine Learning, Introduction to Computing, Advanced Computing, and Data Structures
 - Held weekly recitation sessions and office hours.
 - Produced supplementary lecture materials, prepared and graded programming homeworks.
 - As a coordinating TA, handled all the announcements to the class of 200+ students and worked closely with other assistants to help them prepare and grade homeworks.
- **Sabanci University** Istanbul, Turkey
Moderator Sep 2016 - Jan 2017
 - **Courses:** Calculus 1/2 (MATH 101/2), Natural Sciences 1/2 (NS 101/2), Linear Algebra (MATH201)
 - Prepared content for exam preparation and general review, 6 hours per week.
 - Conducted weekly study sessions to audiences up to 80 students.

HONORS AND AWARDS

- Ranked 1st among batch of 20 students in the Department of Economics at Sabanci University.
- Ranked 5th among batch of 90 students in the Department of Computer Science & Engineering at Sabanci University.
- Dean's High Honor List, 2015-2021.
- Honor scholarship for 5 years, awarded for ranking 1878th out of approximately 2 million people in a nationwide university admission exam.

COMMUNITY & SERVICE

- **Reviewing:** NeurIPS, UniReps, CauScien